Your CRE loan book has blind spots.

I built a system to catch them early — before a payment is missed.

The problem

Hundreds of loans.
One quarterly spreadsheet.
Zero early warning.

Most CRE lenders review borrower financials on a static, point-in-time basis. By the time a covenant breach shows up in a quarterly review, the underlying stress has often been building for months.

What I built instead

01 Synthetic 1,000-loan surveillance database

02 XGBoost + logistic regression risk model, 0–100 score per loan

03 Live Streamlit app for real-time stress testing

04 Power BI dashboard for credit-committee reporting

Why the loan data is synthetic

Real CRE loan tapes are protected by lender NDAs — they can't be published. So I generated a documented synthetic portfolio instead: financially consistent ratios, and default outcomes that actually depend on DSCR, LTV, and occupancy, so a model trained on it learns something real.

100% synthetic loan-level data
0% real borrower data
Full generation logic open-sourced

The uncomfortable result

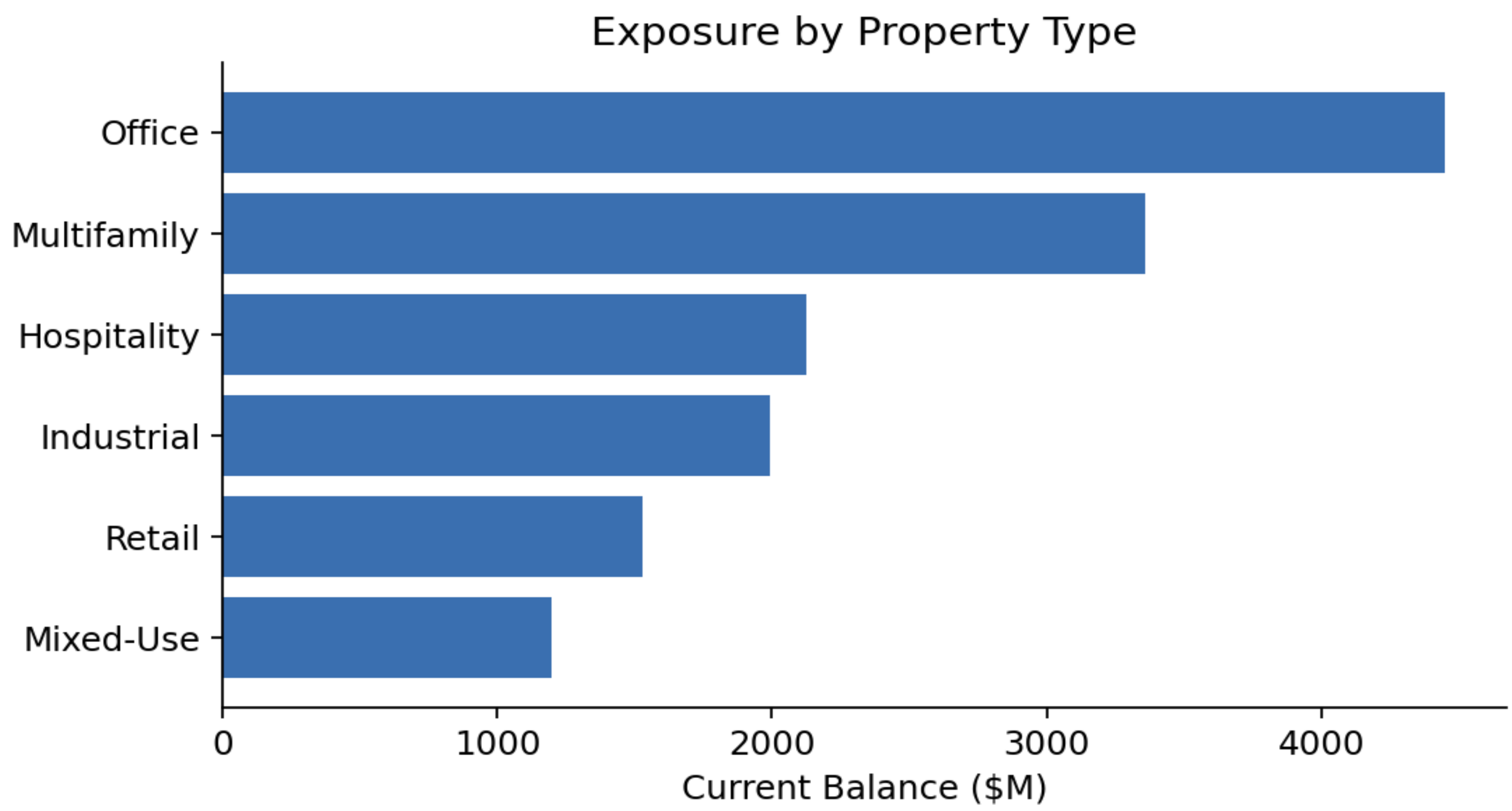
Logistic regression beat XGBoost.

On this dataset size (~1,000 loans, 56 defaults), the simpler model had higher PR-AUC and recall. I shipped that finding as-is instead of quietly picking whichever model looked better — and documented exactly why XGBoost still powers the app's explainability layer.

0.328	**0.285**
Logistic PR-AUC	XGBoost PR-AUC

What the model found

Hospitality loans default 12x more often than multifamily in this portfolio.



18.8% hospitality · 9.8% office · 1.5% multifamily default rate

Try it live.

An interactive risk model you can actually click through — stress-test the portfolio yourself.

Live app: cre-sentinel-debt-surveillance.streamlit.app

GitHub: github.com/emmacg02/cre-sentinel-debt-surveillance